

G-EMV: Emotion and Reason in an Agent. Curiosity, Trust and Commitment

Manel Enrico · Independent researcher, Barcelona · ORCID 0009-0008-1732-6310

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Spanish version, code notes and figures: github.com/Manelenrico/g-emv

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Abstract

This work observes and tests an artificial creature made of two parts. A body that feels what it lacks in three dimensions, its body, its resources and its bonds: what that body feels is what we call emotion here. And beside it a head that talks, a language model that advises it: that is reason. The question is when thinking is of use to a body like this.

To answer it we give the world time, with games twice as long, and we build three pieces. The first is a wide door: the advisor no longer proposes a step, but a whole plan with its emotional shape, what the body will feel on each leg of the way, and the body judges it with its own table. This door tells good advice from random advice, something the earlier one did not do; but, so as not to fool itself, it can only judge the small part of the good that the body is able to foresee. The second is trust, which is earned and lost with results: in the tests, an advisor that talks at random soon loses trust and stops moving the body with good news. The third is curiosity, installed as one more need of the body. As a nudge at each step it did not work. As a trip with a destination and commitment it did: while it travels, the body discovers almost six times more world per instant than when it is not traveling, without getting closer to danger; since it travels little, over the whole game it sees 16% more of the map than the body alone (31% on the stretches where no step was lost; section 10).

In a hundred games, the advisor's plans gave the body no advantage at all. Almost all the improvement in the creature's future goes through the others, and what they will do the body cannot foresee. What the work leaves is the division of roles that makes reason useful: the body says what it needs and, therefore, what is worth knowing; reason can say where to find it; and the body checks it, commits to the plan and only lets go of it if something important happens.

Everything was measured in a single world, private and with chosen neighbors, and without the creature learning from one game to the next.

1. The question

Second part of a story. The first part of this series placed a head that talks beside a body that feels, and asked what it added. The body was the one from three earlier works: an engine with needs on three axes that pull in opposite directions, a table of what hurts, what relieves and what attracts, and a decider that twenty-four times per second imagines a handful of futures, about sixteen, and chooses the one that brings it closest to where it wants to be. The head was a language model that reads the scene and proposes in words. The division of roles has not changed: the advisor proposes and explains; the body imagines, scores and chooses. That first part found that the body heeded the advisor in one proposal out of ten, half of the times to do what it was already going to do, and that the only thing that moved anything was the clock: holding the phrase for four seconds accounted for half of the times the body heeded it.

What was left open. That work ended with a suspicion and a debt. The suspicion was in the door, which is what we call the place through which a proposal from the advisor enters to compete with the body's own. That door turned each proposal into a destination one step away and scored it with the same photo the body uses for everything: where it would be on arriving, with the needs of now and the world frozen. Take a piece of advice like "come out of your hiding place now, on the other side there is a medkit and in a while you will be better". That door could not understand it: it only looked at where the body would be on taking the first step, and on that first step leaving the hiding place is worse. The "in a while you will be better" did not fit, because the photo has no "a while later". So when the body did not take a piece of advice, there was no knowing whether it was because the world did not make use of it or because the door did not let it through. The debt was the test that would have separated the two things: passing through the door a piece of advice already known to be good and seeing whether the body recognized it. Without it, there was no knowing whether the instrument measured anything. And a conjecture remained: that the world, a place where some threat was always on, where nobody recognized anybody and life lasted two minutes, did not give time for thinking to be of any use.

What this work does. It starts with the three things, in order. First it gives the world time: a slower configuration of the same game, with games twice as long, an initial calm and the late ring, without the three players who attack most at the start. Then it runs the missing test, and the earlier door fails it: it always recognizes what the body already wanted and does not tell a future known to be better from any random square. So another one is built. The wide door lets the advisor propose emotional shapes, which is what we will call a plan with a path and with what the body will feel as it travels it: where the body will go, what it will carry, how it will be in life, in resources and in bonds on each leg. The body does not believe the words; it feels each leg with its own table and decides with a rule that first protects life and then looks at what is gained and what is lost. That door does recognize the good future. With it the advisor is given a language to speak

in shapes and a trust that is earned and lost, and it is played in the field. And at the end, with what the field teaches, the body is given a new need, curiosity: to go and see what it does not know. It is a need in the service of survival, because what has not been seen may hide a resource or a danger. And it has an advantage for reason: what the body does not know is something that can be checked, so a plan to go and see it is a plan whose result the body itself can measure.

What was found, in advance. Four things were achieved. The first is an instrument: a door that lets a whole plan in and tells good advice from chance, something the earlier one did not do; so as not to fool itself, it only judges the small part of the good that the body can foresee by itself. The second is a finding about the creature: almost everything that improves its future, nine parts in ten, goes through the others: through where they are and what they do with respect to it, and almost half of that is its sibling. That explains why reason helped so little: what matters is exactly what nobody can foresee. The third is a curious creature: the creature that used to spend the calm looking at nothing now proposes trips to itself toward what it does not know, walks them and discovers six times more world while it travels, without getting closer to danger. The fourth, and the most valuable, is commitment: a body able to hold a plan to the end and to let go of it only if something important happens. And what was not achieved is also said: in a hundred games, the advisor's plans gave the body no advantage at all, and overall curiosity only makes it see a little more of the map, because in this world there is almost never calm. Reason still has nothing useful to propose here; what this work leaves is everything it needs to be able to do so in another world.

Related work. None of the pieces of this work is new on its own. It is worth saying from whom we take each one and where we depart from them.

The body comes from where it came from in the earlier works. The idea of an agent moved by needs that return to their place, as body temperature returns to thirty-seven degrees, and of emotions that choose behavior without anyone writing it, comes from Cañamero (1997). The hypothesis that a machine that looks after a fragile body ends up having motives of its own comes from Man and Damasio (2019).

Curiosity is not new either. Schmidhuber (1991) proposed giving a machine a reward each time it learns to foresee something it could not foresee before, and Oudeyer and Kaplan (2007) put in order the many ways of doing it that came afterward. Active inference, the theory of Friston and his collaborators, goes further: each step is worth what it gives and what it teaches, and the two things add up in a single sum, so curiosity does not need to be added; it comes out on its own (Friston et al., 2015). And the idea that curiosity is better organized in goals than in loose steps also has a precedent: Santucci, Baldassarre and Mirolli (2016) built a robot that discovers goals by itself and decides which ones to practice according to what it learns from them. From all of them we take the underlying idea: not knowing can be a need, and satisfying it is worth something. We depart in three things. Here curiosity is not a reward added to the rest, but one more need of the body, one that threat switches off. The curious trip, for now, does not serve to learn a

skill but to see the world, and for now the world means terrain: squares of the map the body has never seen. It is walked by the same body that flees and eats. And curiosity did not work the first time: as a nudge at each step it was not enough, and it only worked when it became a trip with a destination and commitment (section 8).

Commitment is an idea from philosophy before it is one from engineering. Bratman (1987) held that an intention is a half-finished plan that is not rethought at every moment, and that exactly this makes it useful: whoever revises their plan at every step never gets anywhere. Cohen and Levesque (1990) summed it up in one phrase, intention is choice with commitment, and wrote down when it is reasonable to let it go. Rao and Georgeff (1995) took it into programs that work, the belief, desire and intention agents. From them we take the whole idea: to commit is not to reconsider at every step, and to let go only for a good reason. We depart in where it lives. There, commitment is a rule written in logic. Here it lives in the body: what breaks the trip is what the body feels, a hit, someone armed coming closer, life going down. And here we measure what commitment costs and what it gives.

The advisor that talks follows the line of SayCan (Ahn et al., 2022), already cited in the earlier work: a language model proposes and something that does not talk decides. There, what decides knows what the robot can do; here it knows what the body needs and how it will feel. And this time the model does not propose a step, but a whole plan with the shape of what the body will live through along the way (sections 5 to 7).

We know of no work in which a language model proposes whole plans and those plans have to pass through the consent of a body with needs, which walks them, trusts whoever proposes them according to what they have given it, and lets go of them if they hurt it. That is the proposal. The pieces have owners; the way of putting them together and measuring them is what this work adds.

How it is told. First the body, with what is needed to follow the story and two corrections to what the earlier work said. Then the slow world and what time gives and does not give. Then the two doors, the earlier one and the wide one, and what happened when the advisor spoke through the new one: first on the bench, which is what we call the tests on already recorded moments from earlier games, where nothing moves and everything can be repeated, and then in the field, in real games. Then curiosity. At the end, what remains of the question of when thinking is needed, and how everything was measured. The numbers are in the appendix, each with its source.

2. The body, and two corrections to the earlier work

In short. The body's engine, its table and its way of deciding are the same as in the earlier works and have not been touched. Everything new in this work, the wide door, trust, curiosity and commitment, is added beside it, in pieces that can be switched off; with all of them switched off, the body decides exactly as before. Here we tell just enough to follow the story, and we correct two things that the earlier work got wrong.

A body that feels what it lacks. The creature has needs on three axes: how its body is, how its resources are and how its bonds are. On each axis there are forces that pull in opposite directions, and over the three there is a point where it would like to be. How far it is from that point we will call discomfort. It is a word of convenience, not a diagnosis: discomfort is a distance, and what counts is which way it is going. When it goes down, the creature is better; near zero it is at equilibrium, which is the closest thing to well-being; when it goes up, it is worse. A small discomfort is not suffering: it is being almost in its place. In this world it never gets all the way to zero, because it always lacks something (section 3), but that does not mean it lives badly. Everything the creature does is try to get closer to its place, that is, to make discomfort go down. Between the world and those needs there is a table: a list of situations, each one with the strength with which it pulls and on which axis. Being shot at, the ring coming closer, lacking something in your hands, the sibling being wounded: each thing is a row. When we say that something hurts it or attracts it we say that and nothing more: that a row of its table pulls. We claim nothing about what the creature experiences.

How it decides. Twenty-four times per second, the creature imagines a handful of things it could do, usually about sixteen when it can walk, not all the possible ones: go somewhere, pick something up from the ground, go closer to the sibling, heal itself, stay still. For each one it takes a photo of itself: where it would be and how it would be on finishing. It scores each photo with its table and chooses the one that lowers its discomfort the most. Each of those twenty-four times is an instant; in the game it is called a tick. At the next instant it starts again with another handful. There are no plans, no memory of other games, no reward and no learning: just one step chosen very often. And the world imposes its own rhythm on that step: after moving from one square to another, the legs need eleven instants before they can take the next one. During those eleven instants the creature keeps looking and deciding, but it cannot walk. The legs' pause will appear many times in this work.

The sibling. The creatures play in pairs, and since the third work each one feels its sibling: it hurts when the sibling is hit, when someone has the sibling within range or when the sibling lacks something. Between the two there is a thread, a fixed telegram every two seconds with where each one is, how their life is, what they carry and who is hitting them. It is not a conversation: it is a report.

The advisor. Beside the body there is a language model, which we call the advisor. The situation is described to it in words and it answers with proposals, also in words. The division of roles does not change in the whole work: the advisor proposes and explains; the body imagines, scores and chooses. The advisor does not touch any row of the table and does not decide anything. The only thing it can do is put one more thing among the body's options.

First correction: the creature was not crouching. The earlier work reported that nine in ten times the advisor was consulted, the body was stopped, and from that it drew an image: a prey animal crouching under a threat that does not go away. The figure was correct. The image was not. On looking at why it was stopped, it turned out that in three of every four instants the legs were in their eleven-instant pause, and almost all the decisions to stay fell there. Truly staying still, while able to walk, it chose in only eight instants out of every hundred. The creature was not hidden: it was walking as fast as the world let it. For what the earlier work concluded it changes little, and this must be said: there was still nothing to think about, but not because it was hidden, rather because each decision was "the next step as soon as I can".

Second correction: a rule that did not work. The earlier work said that when a proposal from the advisor tied with what the body was already going to do, the tie went to the body. The rule was written, but because of a programming error it was never applied. We measured it in all the games with an advisor: it affected four out of almost two thousand advisor wins, two in every thousand. No figure in that work changes visibly; what changes is the claim: that rule did not work. The test that should have caught it was looking in the wrong place, and from this comes a rule we follow throughout this work: each thing is checked in the place where it really happens.

3. A world with time

In short. This creature can be given time: a game twice as long, a spell of calm at the start and a ring that arrives late and slowly. With that, the spells without threat multiply by three. But time does not give it a single instant in which it lacks nothing, because what it lacks it carries with it. And with time and without threat, the creature does not start exploring: calm alone does not make it curious.

Why start with the world. The earlier work ended by saying that reason had no place in that world because there was no time: life lasted about two minutes, there was always some threat on and each decision was taken in 40 milliseconds. If that was true, before trying reason again the world had to be given time, or everything else would fail again for the same reason.

The slow world. We do not change the game; we change what the game lets us configure. A game now lasts twice as long: twelve minutes and forty seconds. At the start there are twenty seconds in which nobody can attack; moving is allowed, but any creature that steps off its starting square dies on the spot. The ring, which in the normal game starts closing early, waits until 40% of the game has gone by and then closes more slowly. The aid packages that fall from the sky, which the game calls sponsor, are the same as in the normal game: each pair gets one pod with two rations for one sibling and a medkit for the other. In the slow world they fall at about a minute and a half and at about three minutes, at 11% and 25% of the game, and they land five seconds after being announced. We call this world the slow world, and it is where everything that follows happens.

What time gives. We call calm the instants in which the creature has no fear, its bond with the sibling is at rest and there is nobody armed in sight. In the normal game, calm takes up between three and seven instants out of every hundred. In the slow world, eighteen. Time triples the spells without threat.

What time does not give. If we look for the instants in which it lacks absolutely nothing, with no row of its table pulling, there are none. Not one in almost 40,000 instants lived. And what prevents it is not the ring or the rivals: it is what the creature carries with it. Its hands are almost never full: having at the same time a weapon, a medkit and food, which is what its table considers being well supplied, happens in little more than one instant out of every hundred, and almost always in the same lucky creature. Its life is almost never at its maximum, and the life it is missing hurts it even when nobody is touching it. And if the sibling dies, loneliness stays on for the rest of the game. Figure 1 shows a whole life in the slow world: seven minutes in a row without anyone touching it, with its life full, and not a single instant in which it does not lack something.

A whole life in the slow world · game 20994019 · seat 10
 10,752 instants · nobody touches it until instant 10,273 (seven minutes)
 and at no instant does it stop lacking something

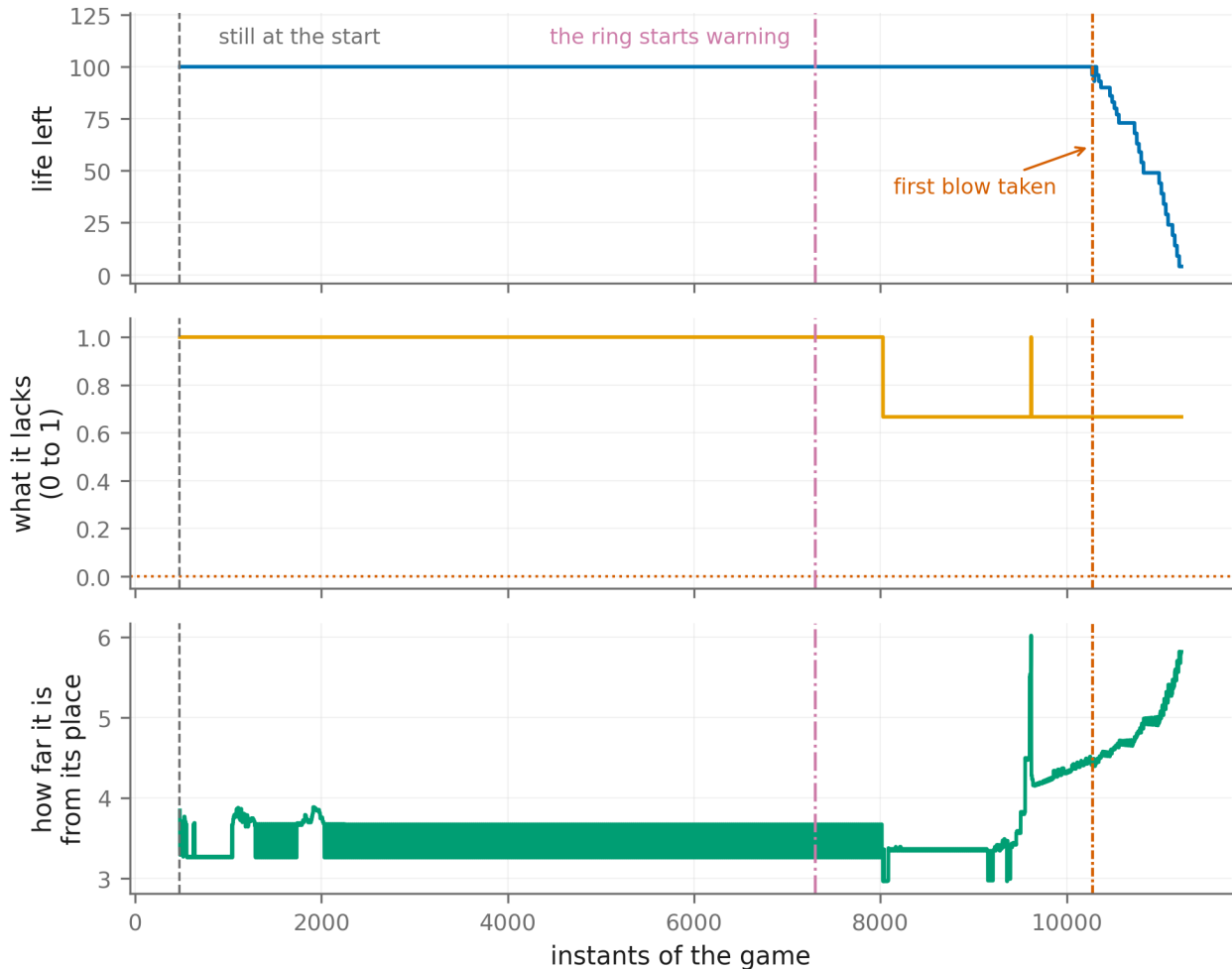


Figure 1. A whole life in the slow world, instant by instant: body alone, game 20994019, seat 10. Top: the life it has left. Nobody touches it until instant 10,273, about seven minutes after the start. Middle: what it lacks, from zero to one. It drops from one to two thirds when it gets something, but it does not reach zero a single time in the whole game. Bottom: how far it is from its place, its discomfort, which rises in the last stretch, when the ring corners it and someone finishes it off. In the central part that line goes up and down so fast that it looks like a solid band: the body goes back and forth between two squares, one step every eleven instants (section 8), and since from one it sees a rival standing still in the distance and from the other it does not, its discomfort goes up and down with each step. It does this for 6,000 instants, from 2,033 to 8,027, the same ones as in figure 3. The vertical lines mark the end of the stillness at the start, the ring's warning and the first blow.

What it does with the calm. With time and without threat, one would expect the creature to start looking around. It does not. In the spells of calm it discovers four new squares for every hundred instants; outside calm, fourteen. It sees three times less world when it is calm than when there is danger. It is not that it prefers to stay still: most of that stillness is the legs' pause we talked about in section 2. It is that it has no reason to go and look at what it does not know. Calm does not produce curiosity on its own. This is noted down for section 8.

Who breaks the calm. On counting who was landing the hits, something came up that we did not expect. In more than 400 lives from the earlier work, in the first 2,000 instants of the game, a little over a minute, three of the fourteen types of rival landed nine in ten hits and caused eight in ten deaths. The other eleven, in that spell, hardly attacked anyone. One of our creatures was shot dead shortly after setting out, with the initial calm at its maximum: calm is not given by the configuration, it is given by the neighbors. That is why, for the rest of the work, we replaced those three with repeats of rivals that hardly attacked at the start. It is a private world, with chosen neighbors, and we declare it as such. And we also declare an error in that choice, found on reviewing the numbers: we only looked at the first minute of the game, and one of the rivals we repeated is not peaceful, but kills late. Counting whole games, it is the fourth deadliest. And in our own games in the slow world it was the one that killed the most: it caused four in ten of the creature's deaths, none before five and a half minutes into the game, close to halfway. Besides, other rivals that kill early remained, and so did the ring. The comparisons in this work are not affected, because all the creatures faced the same rivals in the same places. But the slow world was more violent than we believed when we designed it: it has fewer hunters at the start, not fewer hunters, and what is measured in it does not say how the creature would fare in a truly quiet world.

4. Today's door, measured

In short. This section runs the test that the earlier work left pending, and the result is clear: the earlier door always recognizes what the body already wanted, but does not tell advice known to be good from any random square. What the body accepted from the advisor did not depend on what the advisor said. It was the door that did not let the difference show.

How the test is done. We took forty recorded lives from the earlier work and, every two seconds of each life, one moment: more than 2,200 moments in total. Good advice only exists in the moments in which the creature really was better a while later, which were 908; the figures for that advice are counted over those moments, and the appendix gives the exact number for each one. In each moment we gave the door three different pieces of advice, as if the advisor had said them, and looked at what the body did with each one. The first piece of advice was what the body was already going to do at that moment. The second was advice we know to be good because we took it from the future: the place where the creature was better after a hundred instants, according to its own diary. The third was silly advice: a square chosen at random, as far away as the good one. Each piece of advice had the same chance that advice from the advisor had: for a hundred instants, each time the legs were ready, it competed with the body's options, and it counted as accepted if it won at least once. If what was proposed was exactly what the body was already going to do, that counted as accepted.

What it was already going to do, it always recognizes. The first piece of advice, the one with what the body already wanted, was accepted in every moment. The door never rejects what the body had already decided. It is half good news: it means that the door works as an entrance, but also that what it accepts most easily is what did not need saying.

The good one and the silly one, the same. The good advice was accepted 8.5 times out of every hundred; the silly one, 7. The difference is so small that it cannot be told from chance. And there is something more: those figures are in the same band as those of the real advisor in the earlier work, between nine and eleven out of every hundred, and as those of the voice that proposed places at random, ten and a half. Everything that went through that door was accepted more or less the same, whatever it was. That is why the earlier work could not know whether the advisor was saying good things: the door would not have noticed.

What decided. When the body rejected a piece of advice, the diary notes which row of its table knocked it down. Four in ten times it was exposure: leaving where it was meant being in view. And this happened the same with the good advice as with the silly one. The door judged with the photo of the first step, and on the first step almost any advice that takes the body out of its place is worse, whether what comes after is good or bad.

Distant advice gets through more often than nearby advice. A detail that goes the opposite way from what we thought: the farther away the proposed place was, the more often it was accepted. One step away, two times in every hundred; three steps or more, twenty-three. One step away, the body already had that same square among its options and had ranked it below the one it chose; the advice added nothing, because it proposed something the body had already considered and discarded. Far away, it proposed something the body had not imagined.

There is no word for "stay". In fourteen out of every hundred moments, the best thing the creature could do was not move: a hundred instants later it was better without having taken a step. The right advice was "stay", and the door has no way of saying it: the translator turned it into "go to where you already are", and that was nothing. It is not that the door rejected that advice. It is that it never arrived.

What this establishes. The missing test is now done, and at last it separates the two things the earlier work could not separate. When the body did not take a piece of advice, it was not necessarily because the world did not make use of it: it was because the door could not look beyond the first step. A door that could see what happens next had to be built. That is section 5.

5. The wide door

In short. This section builds the instrument that was missing: a door through which a plan can enter whole and the body can judge it by what will happen, not only by where it ends. That door tells advice known to be good apart from advice given at random, which the earlier door did not do, and it breaks nothing of what the body already did. But in order not to fool itself it can only judge the part of what is good that the body is able to foresee, and that part is small. And in building it the finding appears that orders the rest of the work: almost all the improvement in this creature's future goes through the others.

The test the usual door failed. In section 4 we ran a simple test. We took moments from lives already played and, for each one, we looked in the creature's own diary for where it would have done well to go: the place where, a hundred instants later, it was better off than anywhere else. That is advice we know is good, because we took it from the future. We gave it to the usual door as if the advisor had said it, and we compared it with silly advice: a square chosen at random, just as far away. The door treated them the same. Out of every hundred times, it accepted the good one 8.5 times and the silly one 7. It is like an examiner who gives the same mark to a correct answer and to a made-up one: it is not that the students are bad, it is that the exam does not measure. Before asking the advisor for anything new, the exam had to be fixed.

What a shape is. The usual door judged a piece of advice like a photo: "if I go there, this is how I will be when I arrive". The wide door judges it like a short film. We call that film a shape. A shape is a plan with a path, split into a few legs, like the stages of a trip: first I go to that corner, then I pick up what is on the ground, then I go back next to my sibling. At the end of each leg there is a checkpoint, which is the moment when the body, in its imagination, stops to look at itself in its three dimensions: how its body will be (life, the ring that burns), how its resources will be (what it will be carrying in its hands) and how its bonds will be (the sibling, who will have it within range). And what it feels at each checkpoint it works out with the same usual table, the one of what hurts, relieves and attracts. So a plan stops being a place and becomes a line that goes up and down along the trip, with its three dimensions inside: at this stop I will be worse off in life but better off in resources; at the next one, better off in life and worse off because the sibling is far away. Putting the three together at each stop, as the engine does, gives how far it will be from its place at that moment, its discomfort from section 2. That line goes down on the legs where it will be better off and up on those where it will be worse off, and it is what we call the shape's curve. That is why we also call the shape an emotional shape: what matters about it is not the drawing of the path on the map, but the line of what the body will feel as it travels it.

What it is compared against. A curve on its own says nothing; it has to be compared with something. We compare it with what the body would do if nobody told it anything, which we call going alone. For that, the body also turns into shapes the options it already had at hand at that mo-

ment, about eighteen, and imagines them with the same rules. If the advisor's plan comes out better than the best the body had, it passes. Figure 2 shows a real plan judged this way. That both things are imagined with the same rules is important: the two options are judged by the same yardstick. That does not guarantee that the imagination makes the same mistakes on both sides, because an error can favor one path more than another, and the trap told below is exactly such a case; but it does remove the advantage of judging one with more generous rules than the other.

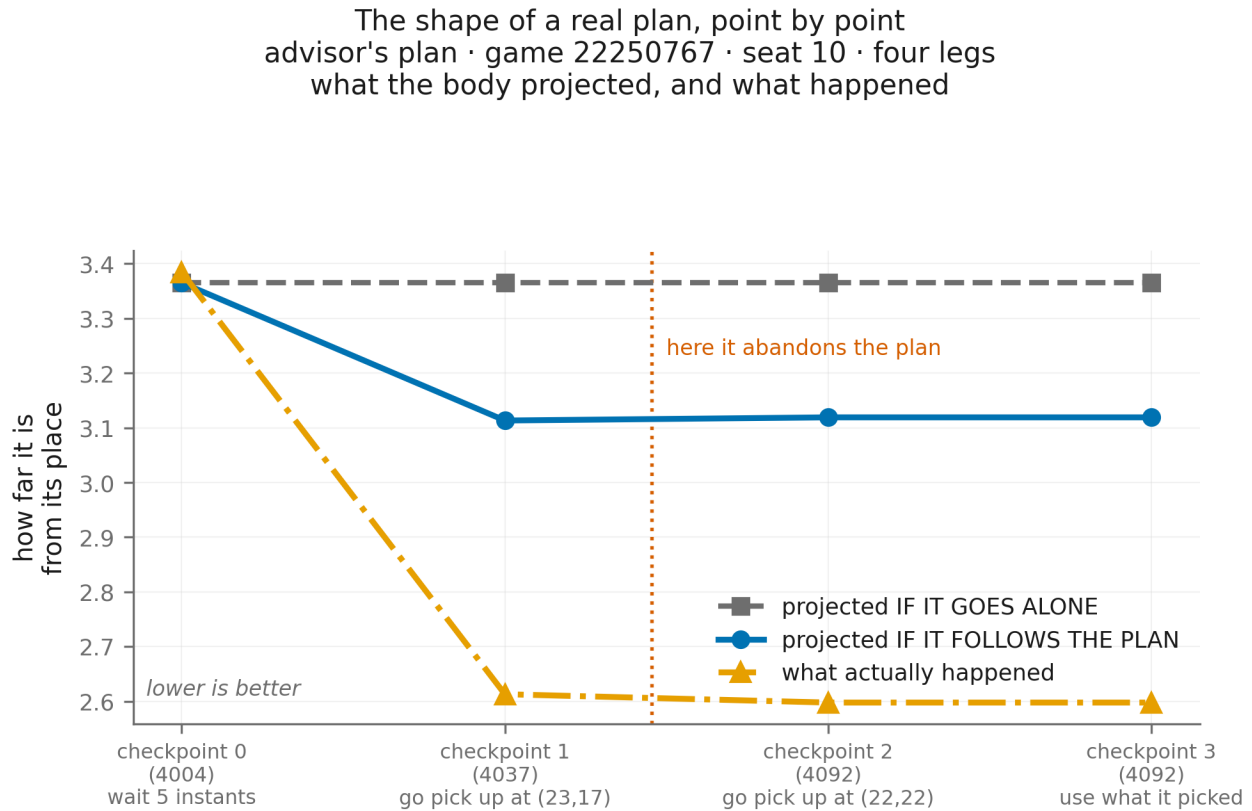


Figure 2. A real plan from the advisor, measured at its four checkpoints: game 22250767, seat 10. The grey line is what the body imagined would happen if it went alone, which at that moment meant staying still, because its legs were in their pause. The blue one is what it imagined if it followed the plan, and the orange one, what really happened. Lower is better. The plan passed the door because it imagined a better path than going alone. The body followed it up to the first checkpoint, and there it was much better off than the plan promised. Almost all of that difference comes from the honesty rule, explained below: when imagining the plan, the body cannot count on a rival leaving, and the rival left. At instant 4,062, on judging it again, the plan no longer beat staying as it was, and the body dropped it: the last two points of the orange line are what it did afterward on its own, and it did not go to pick up what the second leg was for. This is the best case, not the typical one: of the forty-five advisor plans the body started to walk, only ten got to step on the destination of their first leg, and this is one of them.

The rule it decides with. To choose between curves a rule is needed, and this one is designed the way an animal would judge a path. First, life: if at any point on the trip life falls below a minimum, the plan does not pass, however good the ending. Life is not given up for anything. Then, among the plans that do not put life in danger, the good and the bad of the whole path are added up, in the three dimensions together, giving more weight to what is near than to what is far, because the near is more certain than the far. And the plan has to win by a little, not by a thousandth, so that a difference due to noise does not let it pass.

What the imagination gets right. Before judging any plan we checked whether the body knows how to imagine its own future well. We gave it the path it really took and asked it to imagine how it would be a hundred instants later; then we looked at how it really was. When no rival crossed its path, it got it right almost always: in nine in ten of those 180 trips it made no mistake at all: what it carries in its hands, how life goes down if the ring burns it, how it goes up if it is healing. It gets its own part of the future right in most cases. What it cannot imagine is what the others will do to it.

The wide door passes the exam. With the shapes, the rule and the comparison, we repeated the test from section 4 with the same moments and the same good advice. Now the body accepts the good advice seventy-one times out of a hundred, and the silly advice twenty-four points less: a difference too large to be chance. This first version still had a trap, told next, and when it is closed the figures change. And in no case does the new door reject what the body was already going to do on its own: what worked keeps working. This is the positive result of the work, and it is worth saying exactly what it says. It does not say that an advisor will get it right. It says that, if it gets it right, the body now notices.

A trap that had to be closed. At first, random advice also won too often, almost half the time. When we looked at why, a trap came out. In the body's imagination, the rivals stayed still where they were, like statues. And moving away from a statue looks like a safe improvement: if there is someone armed there, any plan that takes me away from them seems better to me. But rivals are not statues; they move, and moving away from where someone was guarantees nothing. So we put in an honesty rule: a plan cannot count as a gain something that depends on where the others will be, because it does not know that; it can only count what gets worse if the path comes closer to someone. With that rule, random advice almost stopped passing, and the few that passed had a real reason: in this world almost any place is closer to something useful than the place where one is standing still, and moving a little has a small real value.

What a good future is made of. With the honesty rule in place, good advice almost never got through anymore: only one in twenty even reached the judgement. Those that get there are still recognized well: out of every hundred that are judged, the body accepts eighty-five good ones and forty-three random ones. That forced us to look at something we had never looked at: why a future is good for this creature. We took almost 800 moments in which the creature really was better off after a while and separated where the improvement came from, row by row of its table. Nine parts in ten went through the others: that the rival no longer had it within range, that it no longer saw it, that the sibling was no longer wounded, that it was no longer being hit. Its own things, what it carries in its hands, what is on the ground, its life and the ring, all together, were one part in ten. In the slow world of section 3 the same happens. Put plainly: what makes things go well for this creature goes almost entirely through how it stands with respect to the others. Part of that it brings about itself by moving, but another part depends on what they do, and that is exactly the part nobody can imagine with certainty; the next part measures it.

What can be known about the others and what cannot. We tried two ways of seeing the others without guessing. The first was to imagine each rival in the worst place it could reach in that time. It is honest, but it does not work: the further ahead in time, the further the rival may have got, and every plan looks more and more dangerous. The second came from the pack: the sibling does not need to be guessed, because it can tell its own plan through the thread that joins them. By letting the sibling tell where it is going to be, the door recovers half of what the sibling brings to a good future. With that, the account stands like this: of what makes a future good, the part that belongs to the body itself it sees whole; the sibling's part, half; and the strangers' part, which is half of everything, it does not see, because to see it one would have to know what they are going to do.

What this establishes. The wide door knows how to see what happens next, and when what happens next is good, it chooses it. That no longer depends on anyone else. But when looking at what makes a future good, we have seen that it is almost always the others: where they will be, what they will do, whether they will shoot or not. And neither the body nor the door can know that; at most one can try to guess. That is where the advisor could bring something the body does not have, because a language model can make guesses about what the others will do. But that is also where it can be wrong, or make things up. That is why what follows has two parts: giving the advisor a way to propose whole plans, and giving the body a way to trust it only to the extent that it has been right before.

6. Trust and the advisor that speaks in shapes

In short. This section prepares the advisor to enter through the wide door, and it achieves two things. The first is a trust that defends the body: on the bench, an advisor that speaks at random about the others quickly loses its trust, and its good news stops moving the body. Its warnings of danger, on the other hand, are still believed, out of caution. The second is a language for the advisor to propose whole plans, and with it the advisor goes from getting not a single piece of advice through to getting one in ten through. What it does not yet achieve is for the advisor to know the body: it learns to speak the language, but it still does not know well what the body will feel with what it proposes. They have only just met, and in this work the advisor has no way to learn from one game to the next.

Why trust is needed. Section 5 ended by saying that what makes the creature's future good goes through the others, and that the part that depends on what they do can only be guessed. If the body is going to listen to someone who guesses, it needs a way of knowing how much to rely on them. And it needs it for a more serious reason: as soon as the body scores what someone promises it, words can move it, and with them errors too, whether unintended or not. Trust is the defense. Here trust is understood as the first bond, the one with oneself, between the body and its head, and it goes in both directions. The body keeps count of how much the advisor's promises resembled what happened afterward; the advisor, of how much notice was taken of it.

Three test advisors. Before trying with a real advisor, we tried trust with three made-up advisors, which are not the language model but simple programs, that say things about where the rivals will be at each stop of a plan. The first tells the truth, as far as it can be known: the real position of each rival if either of the two siblings saw it at that moment. The second speaks at random: it gives made-up positions, without looking at anything. The third always says the same thing: that nobody moves. What the advisor says only enters the body's imagination to the extent that it has earned it, with one exception: the bad is always believed whole, because in this body a warning of danger is not discounted.

Trust defends. The one that speaks at random loses the body's trust in a few stops and no longer manages to get a single plan through. This is measured without psychology: it is enough to compare what it said with what the body saw afterward.

But why trust is given matters. If trust is earned by being right, the one that says nobody moves earns it just as the one that tells the truth does, because in the time a plan lasts the rivals hardly move: being right is easy without knowing anything. If it is earned only by getting right what is new, that someone arrives or that someone leaves, the one that says nobody moves stays at zero and only the one that sees the changes coming earns it. The way trust is handed out decides who is listened to. And there is a detail worth keeping in mind: trusting halfway is the worst. With half

trust, the door lets random plans through without giving the good ones enough; the word has to enter whole or not enter at all.

The real advisor knows no more than the body about the others. With all that prepared, we went over what the real advisor of the previous work had said in more than 3,000 answers. It talks about the others eight in ten times, but almost always to repeat where they are, which is what it had already been told. When it dares to announce a change, that someone is going to arrive or that someone is going to leave, it is right less often than the assumption that everything stays the same: half the time against almost nine in ten. So trust could not be given to it for its predictions about the others. It is given for what happens after following it: if the body, as it walks the plan, is as it imagined or better, trust goes up; if it is worse, it goes down.

The language of shapes. What remained was to give the advisor a way to propose whole plans. It is the most original piece of this part. A plan is expressed as a sentence, and each leg of the plan is a clause with two halves. The world half says where to go and what for: to that corner, for the medkit, next to the sibling. The emotional half says how the body will feel on that leg, in the three dimensions of the model: whether life will go better, the same or worse; whether the hands will fill or empty; whether the bond with the sibling will tighten or loosen; and which row of the table causes it. The grammar of that language is the three dimensions, and its vocabulary, the rows of the table. The body does not believe the emotional half: it imagines each leg with its own table and puts in its own numbers. But that half tells the advisor what it has to think about.

From zero to one in ten. We tested it with the same language model as in the previous work, on 200 recorded moments, with the same description of the world and the same instructions, changing only what it was asked for. Asking it for single steps, as in the previous work, the wide door accepted none of 367. Asking it for plans in the language of shapes, one in ten. With a larger model, almost two in ten, but in the competition with the body's options the two models ended up the same. What opens the door is not a larger model: it is the language.

It speaks the language, but it does not yet know the body. The advisor learned the grammar right away: almost all its legs had both halves, it used almost all the rows of the table, and it always answered in the format. But what it said the body would feel was seldom right. For the hands it got the sign right two times in ten, less than choosing at random among the three possible answers: it said "the hands will improve" when going toward an object, when in this body's table the lack is only calmed with the weapon already in hand, and the attraction of the ground goes up when getting closer. It gave the sign of desire, not that of need. Where it was right most often was the bond, six in ten, and that is the only dimension about which it was given a real piece of data, what the sibling tells through the thread. What it is given as fact it uses well; what it has to deduce from how the body is made, not yet. It is not that it cannot understand it: it is that nobody has taught it and it has not had time to learn it. In this body there are things that calm down little by little as one gets closer and others that only calm down when one has them in

hand, and that cannot be guessed from outside. That is why, for the field, we gave it two aids. We explained to it in one sentence each row that is not intuitive. And in each query we told it how its previous plans had gone: what it said the body would feel, what the body imagined and what it really felt. Whether that was enough for it to learn within a single game is told in section 7.

Two more things. Asking it not to guess where the rivals would be worked: with plans, it named them in one answer in seven, whereas with single steps it named them in seven in ten. And it still almost never kept quiet, even though it was given permission: once or twice in a hundred.

What this establishes. Trust is a defense that works, as long as it is given for what happens and not for easy predictions. The language of shapes lets the advisor in where the language of steps never did. And the advisor knows how to say where to go, which the body can check, but it does not yet know how to say what the body will feel: it still has to get to know it. With that, everything was ready for the field, which is section 7.

7. In the field: the advisor that speaks in shapes

In short. In a hundred games, the wide door really works: it does not put the body in danger, there is no sign that it makes it miss good opportunities, and it lets the advisor in. The body walks the plans it accepts, and the advisor learns something new: to keep quiet when it has nothing to say. What does not happen is that its plans help: in this world they give the body no advantage.

What was tested. The bench said that the wide door lets the advisor in. What was missing was the usual thing: knowing what happens in the real world, with the clock running and the neighbors moving. We played a hundred games in the slow world of section 3, with three creatures on the same maps and the same rivals in the same places, so that the only thing that changed from one to another was what they carried inside. The first was the body alone, the usual one. The second carried the advisor speaking in shapes, with the body's table explained in its instructions and with a report that, at each query, told it how its previous plans had gone: what it said the body would feel, what the body imagined and what it really felt. The third received random plans, of the same length and the same reach as the advisor's, with nobody thinking behind them. That third creature is the control the previous work taught us: a plan can win just because it lasts, without anyone having thought anything, and one has to know how much of what the advisor wins is that. All three went through the same honest door of section 5, and in all three the body kept count of trust: it went up when what happened was not worse than what had been imagined, and went down when it was.

The machine. Two hundred lives with the whole diary, 1.5 million instants in which the body decided every time, without skipping any, and more than 2,000 queries to the advisor, all with their report. Imagining a whole plan takes longer than one instant of this world, so it is done separately, in parallel, when the plan arrives and then every twenty-five instants. What we did not see then is that preparing that separate work took time away from the body, and some of its steps did not reach the world; we tell it in section 10, with what it changes and what it does not. There was also another failure, one we did catch in time: the report that was supposed to tell the advisor how things had gone arrived empty, and the check that was supposed to catch it was looking where the report was not. It was fixed, it was checked by looking at what the advisor had really received, and those first games were played again.

The door works live. No advisor plan that the body accepted took it below the life minimum. And in the more than 3,000 plans the door rejected, the body alone, at those same moments, was not better off afterward either. That does not show that a rejected plan would not have been useful, because we do not know what would have happened by walking it; what it says is that at those moments there was no improvement at hand that the body alone took advantage of, and that there is no sign that the door threw away good opportunities. The advisor gets in, a little but it gets in: two or three plans accepted out of every hundred it proposes. On the bench, that same

door had accepted none of the 367 single steps the advisor proposed with the language of the previous work (section 6).

The body walked the plans. Before judging the advisor's plans a suspicion had to be ruled out: that the body accepted them and then did not follow them, that between one step and the next it got distracted and went back to its own business, as would later happen to curiosity in section 8. It was not so. When the body had an accepted advisor plan and its legs ready to take a step, three times in four the step it took was the plan's. The advisor's plans were short, two or three squares, and a short plan holds up almost on its own. So what comes next is about the plans, not about a body that did not follow them.

What gets in does not help. To know whether a plan was good we looked at what happened after accepting it. At each stop of the trip we compared how the body really was with how it had imagined it would be. If it was the same or better, the plan had kept its promise. The advisor's plans kept it at forty-two out of every hundred stops; the random plans that passed the same door, at sixty-one. The advisor comes out worse than chance. The difference is right at the edge of what can be chance, and it also has to be qualified: the random plans were shorter and closer, and a short plan promises little and keeps its promise easily. Comparing only similar plans, the difference drops by half and can no longer be told apart from chance. And there is another measure, a simpler one: whether after accepting a plan the body ended up closer to its place, that is, whether its discomfort went down. It did not go down: in the typical case it stayed exactly where it was. The body alone, at those same moments, did not lower it either. This means the advisor proposed in quiet moments, when there was not much to gain, and its plans gained nothing. And at the end of the game, the creature with the advisor lived just as long and finished in the same position as the creature alone.

What changed in the advisor, and what did not. One thing did change, and it is new in this series: the advisor learned to keep quiet. In the previous work, with permission to say nothing, it did not keep quiet even once. Here, with the report in front of it and the body's table explained, it kept quiet four in ten times it was queried, and it kept quiet more when the body's trust in it was low. It never kept quiet before any of its plans had been judged: it kept quiet when it already knew how things had gone for it. What the report did not teach it was to understand the body. Each advisor plan says, leg by leg, how it thinks the body will feel, and those predictions were right just as often at the end of the games as at the beginning: the report made it speak less, not be right more. Within a game, with its few dozen queries and no memory of earlier games, it did not get to know the body. Whether it would manage with more time together, remembering from one game to the next or training on what the body felt, is a question that remains open. And three things that were built for this series never came into play, because all of them depended on the door accepting plans often, and it accepted few: the sibling telling its own plan through the thread, which did not change any decision; the body silencing the advisor when trust hit bottom,

which never happened; and trust itself, which ended the games at the same point where it started them.

What this establishes. It is what section 5 announced. In this world, almost everything that improves a future goes through the others; an honest plan can only count on the small part that belongs to the body itself; the advisor cannot guess the others better than assuming nobody moves; and what it has left to propose, going toward something on the ground or toward the sibling in a quiet moment, is what the body already does on its own. The wide door sees what happens next, is not fooled by still rivals and did not put the body in danger; the language of shapes lets the advisor in; and the body walks the plans it accepts. What fails is that the advisor, here, has nothing useful to propose that the body can check. It is not that reason is of no use: it is that this world does not give it a job the body can verify. Section 8 tries to give it one.

8. Curiosity

In short. A creature that looked at nothing during calm learns to go out and look. Curiosity works when it stops being a small nudge at each step and becomes a trip that the body proposes to itself and commits to: while it travels it discovers almost six times more world per instant than when it does not travel, without getting closer to danger and without any cost to its life that we can detect. Because it travels little, over the whole game it sees 16% more of the map than the body alone (31% on the stretches where no step was lost; section 10). How much it explores in total is decided by the world, because it only travels when it is calm. And what had to be built for it to work, a body able to hold to a plan, is the piece that reason needed.

8.1 A body that does not look

Up to here the paper has told how reason touches emotion: through a door, which we have made wide so that plans fit through and not only steps, and with a trust that is earned and lost. What came out is that the door works but what comes in through it does not help, because almost everything that is worth something in this world goes through the others, and the advisor cannot guess what they will do. We had one question left that was not about the door but about the body: if the body is given time without danger, does it do anything with it?

The answer is measured over a whole life of the slow world: between instant 1,977 and instant 8,027 the threat is zero, more than 6,000 instants of calm in a row, and the fraction of the map that the body has never seen stays stuck at the same number (figure 3). The body is not still: it goes back and forth between two squares, one step every eleven instants, caught between two pulls that take turns. From one square some arrows nearby pull at it, and it takes a step toward them; from the other, that option no longer comes up and the loot pulls at it, which makes it go back. The arrows stay there for the 6,000 instants and it never picks them up. It is the defect we will see further on with the trips: a body that decides everything again at each step holds on to nothing. And it does not look. Calm does not produce exploration by itself, just as an animal that is fed and safe has no reason to get up. If we want it to look, we have to give it a reason to look, and that reason is curiosity.

Six thousand calm instants without looking at the world
 game 20994019 · seat 10 · body alone
 no threat at all and not one new square discovered,
 from instant 1,977 to 8,027

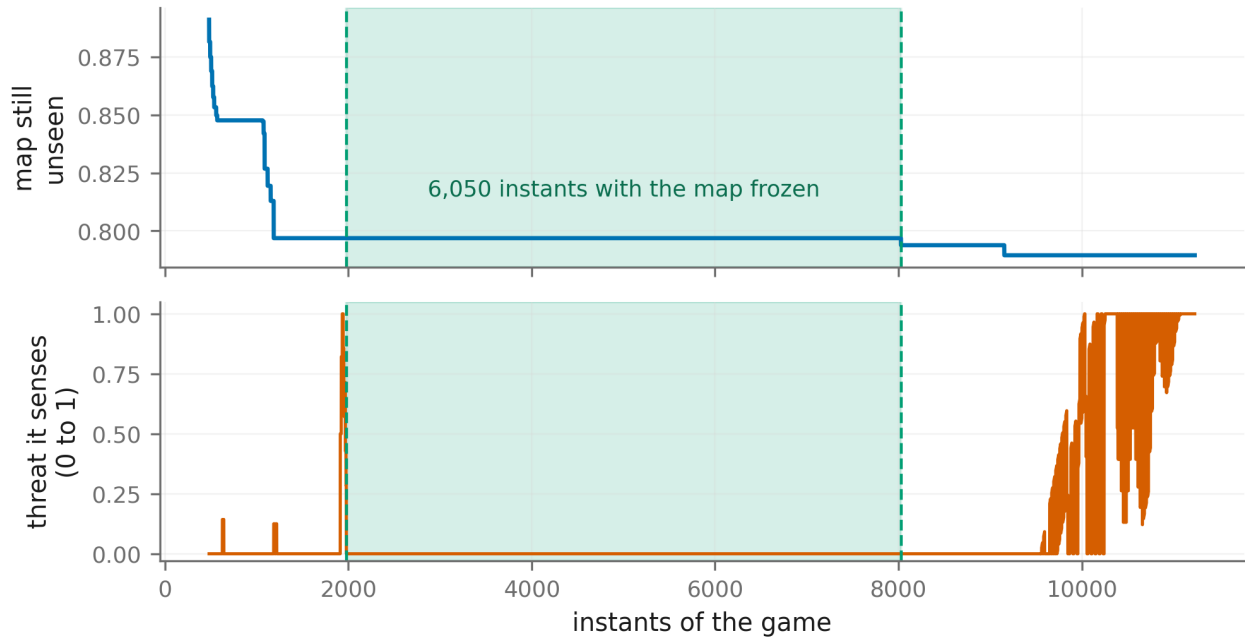


Figure 3. The same body as in figure 1, seen from another side. Top: the part of the map it still has to see. Bottom: the threat it senses, from zero to one. In the green band, from instant 1,977 to 8,027, 6,050 instants in a row, the threat is zero and the map does not move: not one new square. The body is safe and it is not still, since it goes back and forth between two squares, but it does not explore. That is the gap curiosity comes to fill.

8.2 Curiosity as a need

We installed it as one more need, of the same kind as the lack of a weapon or a medkit: what has not been seen of the world is a lack, and getting closer to it relieves that lack. The body keeps a map of what it has seen; what it has not seen is its ignorance, and that number is its own, as much its own as its life or its hands. This matters, and for a reason that connects with everything before. The advisor could not help because what was valuable depended on the others, and what the others will do nobody knows: any plan that promised something about them was a bet that the body could not check ahead of time. With ignorance the opposite happens. Which part of the map the body has seen and which it has not is something the body itself knows exactly. A plan that says "go over there and you will see what you have not seen" makes a promise that the body can calculate beforehand and check afterwards. For the first time, reason has something to propose that the body can measure with its own accounts.

And it carries a balance: the same need weighs differently depending on whether the surroundings are safe, neutral or unsafe. The threat is a number between zero and one that the body calculates from what it sees: someone armed nearby, a recent hit, the ring. Curiosity is multiplied by

one minus that threat, so that the more danger the body sees, the less curiosity weighs, and it only falls completely silent when the threat is at its maximum. With someone armed within range, moreover, the body's usual veto decides, and curiosity does not count. That the balance is not decoration is checked by taking it away: without it, curiosity changes decisions in unsafe surroundings ten times more.

Installed this way, curiosity is a pull: at each decision it pushes a little toward the step that gets closer to the unseen, just as hunger pushes toward food. As a one-step pull, it did not work. In forty games the curious body did not explore faster than the body alone: the new squares per hundred instants were the same. And the reason was not distance, as we first thought: the unseen is almost always six squares away, behind a corner, because the body's vision hides half of what is around it. The reason was the step by step. The body can only decide one step in every eleven instants, because the legs take time to cool down, and a pull that, even in safe surroundings, changes one decision in eighteen takes nobody anywhere: the body takes a step toward the corner and at the next one goes back to the plate.

8.3 Curiosity as a shape

Then curiosity stopped being a pull and became a shape: a whole trip, "I'm going to that corner and I'll see things along the way", which the body proposes to itself and which the same wide door of the paper judges as it judged the advisor's shapes. It is the first shape that the body proposes on its own, and it joins the two halves of this paper. Let us recall the honesty rule from section 5: when the wide door examines a plan, only what the body can foresee by itself counts as gain; what depends on the others does not count, because nobody knows what they will do. Almost nothing the advisor proposed passed that filter. Curiosity does pass it. How much new map the body will see when it reaches the corner it can calculate on its own, looking at its own map, without guessing anyone. That is why the relief of no longer not knowing does count at the door, and it shows: if the door does not count it, it lets through three trips in a hundred; if it counts it, one in three.

It took a few preliminary test games before it worked, and each one found a defect that was not in curiosity but in the body. The shapes died at twenty-five instants because we examined them with a clock of instants in a body that walks at eleven instants per square: the examination fell a third of the way along. We changed the examination so that it would not arrive before the body could have arrived, counting the way in steps and not in instants; the shapes lived, but they were not completed, and the reason turned out to be the most important of the section: the body did not follow its own plans. Between stride and stride it argued everything again, and half the time going for something that hurt less at that instant won. A body like that cannot hold to a long trip, neither its own nor the advisor's. The advisor's plans were short, two or three squares, and those held almost by themselves (section 7); a curiosity trip, of four to eight, did not.

8.4 Commitment

The solution is one of design and it is simple to say: when the body accepts a trip, the trip becomes a goal and is walked to the end. Only something important breaks it: someone armed coming toward it, a hit, life going down, something lost from its hands, or the clock's examination saying that it is no longer worth it. Seeing someone with a weapon in the distance is not important; the body goes around them. Someone coming toward it is.

With commitment, the body obeys its own plan nine strides in ten, even when its instinct prefers something else, and that costs it something. Holding to a plan when instinct pulls the other way is an effort, and the effort shows in the discomfort: a small fraction for each instant in which it obeys the plan against its instinct. It is what committing costs, and it is worth it: over the whole game it does not show, because the curious body lives as long as the body alone (subsection 8.5). And for the first time a curiosity trip keeps its promise (figure 4): seven in ten are completed, that is, the body gets to see what the trip promised to show it. It does not step on the destination in any of the 209 trips of the forty games, and it does not need to: the square it was heading for can be seen before reaching it, and seeing is what it was after. A trip for looking is fulfilled by the eyes, not by the feet. Inside a trip it discovers almost six times more world per instant than outside, in all twenty seeds without exception (a seed is the number that fixes the map and the rivals of a game; with the same seed, the two creatures start the same game: the same map and the same rivals in the same places). Commitment is not blindness: in the forty games it was broken for five different causes, and the curious body gets closer to armed rivals less than the body alone.

A life of the curious body, with its 18 trips
game 20260916 · seat 11 · each band is a trip, coloured by how it ended

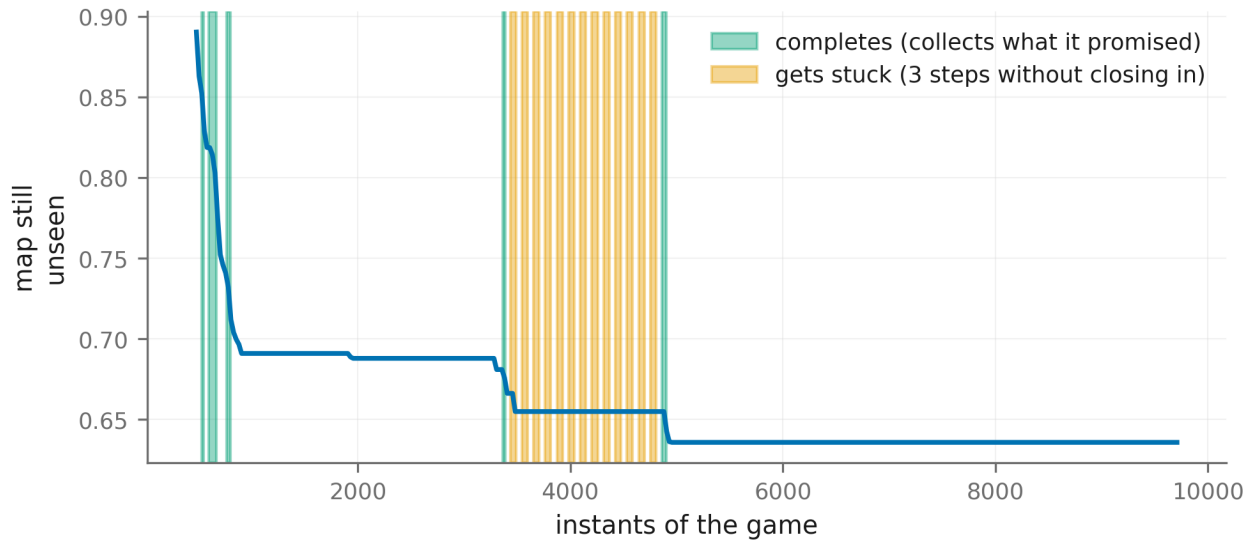


Figure 4. A life of the curious body with the eighteen trips it set out on: game 20260916, seat 11. The blue line is the part of the map it still has to see; each band is a trip. In green, the ones that are completed: the body sees what the trip promised, and the map drops. In orange, the ones that get stuck: the body takes three steps in a row without getting closer to the destination and the trip closes. The thirteen orange ones in the middle all go to the same place: the body proposes the same trip thirteen times in a row, and all thirteen times it gets stuck. After instant 5,000 it sets out on no more: for more than 3,000 instants the threat hardly ever drops enough for a trip to be born, and the seventeen born after that do not pass the door. This is the life with the most trips in the forty games, not the typical one: here five of eighteen are completed, and overall, seven in ten.

8.5 What decides how much is explored

And yet, one of our predictions failed. In this work we write each prediction before playing the games, with the figure that would have to come out to accept it, and we call it a seal because it is closed before looking and can no longer be changed. The seal asked for the curious body to see half again as much map as the body alone, and it failed. It sees 16% more, and the figure holds whether ten, fifteen or twenty seeds are counted (figure 5). The arithmetic explains it: the shape is alive 2% of the body's life. A body that explores intensely 2% of its time explores a little more in total, not much more.

How much of the world each body gets to know
 40 lives of the body alone against 39 of the curious body · the forty games
 (thin lines: each life · thick line: the median)

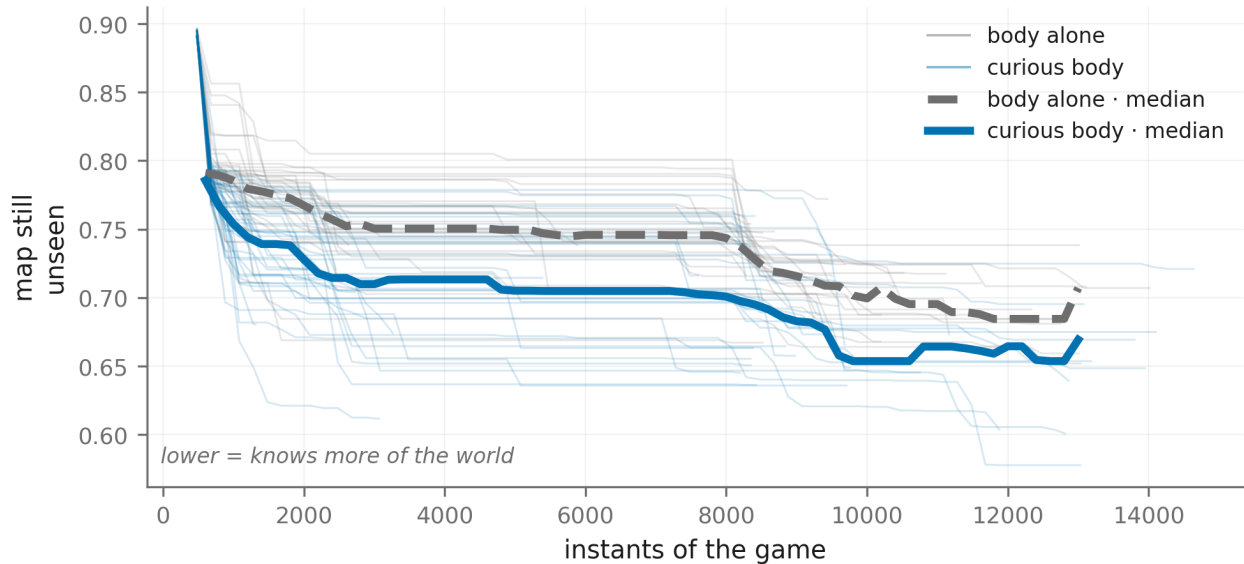


Figure 5. The seventy-nine lives of the forty games: forty of the body alone and thirty-nine of the curious body, because the diary of one never arrived. Each thin line is the part of the map one life still has to see. The thick ones are each group's median, the life that sits in the middle if they are ordered from most to least, and they are drawn only while at least five lives of that age remain. Lower means knowing more of the world. The curious body's median stays below the body alone's for the whole game, but the distance is small: 16% more world, not the half again the seal asked for. The rises at the end are not forgetting: as fewer lives remain, the median is made up of different ones.

What fixes that 2% is not curiosity, it is the world. A shape is born every forty-three instants of safe surroundings, and that ratio barely varies between seeds, while the number of trips born in a game goes from ten to almost 400, depending on the safe time it happens to get. What the body has left to see is the same in all games; what changes is how long it is calm. It is the hierarchy of the model measured in the field: curiosity lives above survival, and in a world where the loot never stops calling and the rivals never stop appearing, the upper floor has little room. There were two ways to give it more: lower the bar of the door so that longer trips got in, or let them pass near the armed ones. We tried the first, and it gave trips that all died at the sight of the first spearman. The second we did not want: letting a curiosity trip pass near someone armed is risking life to see a little more map, and that goes against the rule of the whole model, which first protects life and then looks at the rest. It is exactly what the balance prevents. The lever that remains is outside the body: a world with less loot and more calm.

We do not detect that curiosity costs the body anything over the whole game: in the forty games, the curious body lives as long and ends in the same place as the body alone, with no difference that could not be due to chance. A small difference could stay hidden in the noise, and that is said in section 10. There are two other predictions that failed, and both because of how they were

measured, not because of what the creature did. The first asked that there almost never be someone armed within range while a trip was still under way. It happened more than foreseen, but in those instants the trip was not in command: commitment only commands when the legs are ready to take a step, and with someone armed within range the body breaks the trip and decides with all its rows and its usual veto. The prediction counted instants of exposure that the body handled well. The second asked that thinking the trip should not cost more than 5 milliseconds per instant. One of the thirty-nine creatures measured went over that limit, with 5.51 milliseconds, and it went over it in all its instants, also in those in which it was not thinking any trip: it was slow in itself, not because of the trips. And it did not skip a single decision, although this measure could not see whether the steps reached the world, and not all of them did (section 10).

8.6 Curiosity about the others

After curiosity about the terrain came curiosity about the others: getting to know whoever is not known, and having that knowledge carry a sign. In the model, a positive bond is the root of pleasure, joy and the wish to be close; a negative one, the root of displeasure and, when the surroundings stop being safe, of fear: fear of pain and, at bottom, of losing life. Fear makes one move away from whoever causes it and leave everything else for later. In this first version the bond carries only the sign, good or bad, and does not yet distinguish between the emotions of the same sign. We added it as two rows and tested them on recorded games, measuring over 434 diaries.

The result goes the opposite way from what we sealed. We thought the creature would die at the hands of those it did not know; in 418 deaths there was not one at the hands of someone unknown, someone the body had never seen. It died at the hands of rivals it already knew almost completely. There is a simple explanation. Deaths are caused by rivals that attack, and those, when they kill, the body already knows almost completely: in the typical death it knows more than nine tenths of what can be known about someone about its killer. But knowing someone, in the sense in which the body measures it here, is knowing where they have been and what they have done; it is not knowing when they are going to shoot. The danger did not come from what the creature did not know, but from what it knew and could not avoid. And since the body gets to know the others in the first 2,000 instants and after that there is nothing new to know about them, curiosity about getting to know them runs out soon. What stays alive is the bond with a sign, and it had a defect. On curiosity about the terrain we had put the balance: fall silent when there is danger. On the bond we did not put it, and that is why it came to change some decision of the body with someone armed having it within range, one or two depending on the strength with which it was installed, which is the moment when no curiosity should have a say. It is an oversight of design and not of conduct, and the next version corrects it by giving the bond the same balance.

8.7 What those who minimize surprise would say

Whoever works with active inference will recognize here their epistemic term: a plan that is judged not only by what it achieves but also by what it teaches. We grant it, and we add where we do not agree. First, that in our body exploring is not built in: without installing it, more than 6,000 instants of calm with the ignorance frozen. Second, that curiosity is not added to survival in a single account where a lot of information can make up for a little risk: it goes underneath. With curiosity as a pull, in more than 27,000 decisions with an armed rival within range none was one of curiosity; with curiosity as a trip, when someone armed appears within range the trip breaks and the whole body decides with its veto. And the balance, without which the body walks into danger, shows it by subtraction. In active inference, risk and information are added in the same account, so that, depending on how the preferences are written, a large reward of information can make up for some risk. Here there is no common account: in danger, curiosity weighs less, and within range it does not decide. Third, commitment: a plan without it comes apart between stride and stride, and we had to give the body the ability to hold to it and a short list of things that break it. And in the background, what we already said in the previous paper: for them surprise is the foundation; for this model it is an episode. For active inference, a living being is above all someone who tries to keep the world from surprising it, and eating, fleeing or exploring are ways of achieving that. For this model, a living being is above all someone who lacks something: life, things in its hands, company. Eating, fleeing or exploring are ways of getting closer to where it would like to be. They would say that this very thing fits in their account, because they write needs as what the body expects to find, and they are right that the accounts can be translated. The difference is not in the accounts, but in what is put on the ground: here surprise arrives, lasts a moment and goes away, while the lack is always there. The body is not made of surprise, but of what it lacks.

8.8 What the paper takes away

Curiosity fits in the body, it is cheap, and we do not detect that it does it harm. As a shape with commitment, it does what was asked of it while it acts; how much it acts is decided by the world. And what is worth most in this section is not curiosity but what had to be built for it to work: a body that accepts a plan and walks it to the end unless something important happens. That is the piece reason was missing in order to steer. With it, the questions of the next work, which section 9 sets out, now have ground to stand on.

9. When thinking is needed, second round

In short. The previous work said that thinking is needed when there is time. It was right, but that was not all: time is needed and three more things besides. A job that the body can check, a body able to hold to a plan, and a world with calm stretches. The first two are already built. The third is given by the world.

Where thinking fits. No plan fits in two minutes of life, so time was needed. But this work gave time, a door that sees what happens next and a language to speak in plans, and even so reason did not help. The reason is that in this world what makes the creature's future good goes through the others, and what the others will do nobody knows. Any plan that promises something about them is a bet that the body cannot verify ahead of time, and the advisor, when it bets, is right less often than assuming that nothing changes. Thinking is useful when there is something to gain by looking further than the next step, and that something has to be measurable from inside the body. The idea is similar to that of Lisa Feldman Barrett (2017): for her, the first job of a brain is to keep the body's accounts, to get ahead of what it is going to need before it runs short. She tells it in the language of prediction, which we do not use here, but the core is the same: the head works for the body. And a plan from the head is only useful to the body if the body itself can notice, with what it itself feels, whether it went well: whether life goes up, whether the hands fill, whether the sibling is still close. When the head promises something that the body cannot foresee by itself, such as what the others will do, the body has no way of knowing beforehand whether it is good for it to listen; it can check afterwards, but afterwards comes too late to decide. That explains what happened to the advisor in this work. The advisor worked: it spoke in plans that the door understood, and the body walked them. What it lacked was something to offer that the body could foresee and check on its own, because in this world almost everything that was worth something depended on the others. Curiosity shows the way: how much new map it sees on arriving, the body notices on its own. The next work gives the advisor that material: the sibling, who speaks and can be known, and clear goals with time to meet them.

The first piece: a checkable job. Curiosity gave the first example. Which part of the world the body has seen is something the body itself knows exactly, so a plan to go and see what it has not seen promises something that the body can calculate and afterwards claim. It is not the only task like that. The state of the sibling, who speaks, is another: it is the only part of the others that can be known without guessing. Reason has a place where its promises fall within what the body can check.

The second piece: holding to a plan. A plan is of no use if the body abandons it between one step and the next. Curiosity showed this bluntly: without commitment, the body accepted the trip and at the next stride argued everything again. With commitment, it walks it to the end and only lets go if something important happens. This piece does not belong to curiosity: it serves for any

plan, whether it comes from inside or from reason. It is what reason was missing in order to be able to steer.

The third piece: a world with calm. Curiosity explores six times more while it travels, but it only travels when it is calm: it sets out on a trip once every forty-three instants without threat, and in this world those instants are few. The world was always calling and always threatening: the loot on the ground never stops calling, and a rival we thought peaceful caused four in ten deaths, all after five and a half minutes of the game (section 3). The rivals can be changed without touching the game, by choosing them from whole games and not only the first minute. The loot cannot yet: the game has no setting for how much loot there is, so that would need a change to the game. That leaves within reach a world with more calm for the next work.

What it opens. The next work has two questions, and both come from here. The first is whether a reasoner can lead the body toward goals that the body does not seek on its own, with the body accepting or rejecting each plan with its own door and without coming off worse: a small, measurable form of aligning a reasoner with a body that feels. It is an intermediate step: before aligning a reasoner with us, one can learn to do it with a body whose needs can be read in full. In that division of roles, the body says what it needs and, therefore, what is worth knowing; reason can say where to find it, and the body checks it. The second question comes from the finding of section 5: if the only other that can be known is the sibling, because it speaks, then the natural place for reason is the pair. Two bodies, each with its head, that speak to each other through their bodies and agree on a plan: you keep watch while I gather. And beneath the two lies a third: that the advisor gets to know the body it advises. Here they met anew in each game. With memory from one game to another, or learning from what the body felt after each plan, it could refine its language and its advice. Its shapes already say how the body will feel along the way; what it lacks is getting it right, until what it promises the body will feel resembles more and more what the body really feels on arriving.

10. How it was measured and how far it reaches

In short. Everything this work says comes from games actually played or from tests on recorded moments, and each sentence says which. The predictions were written before measuring. The body was not touched. And the limits are stated here in full.

Two ways of measuring. The bench is tests on moments already recorded from earlier games: a moment is taken, the door is given a piece of advice, and we look at what the body does, without the world moving. It is useful for comparing many things on the same moments, but it does not say what would have happened afterwards. The field is real games, on the game's platform, with the rivals moving. Nothing measured on the bench is claimed for the field without having measured it there; the rule was learned the hard way: on the bench, the decisions of curiosity as a pull promised dozens of times more new squares than those of the body alone, and in the field the curious body saw 9% more.

Where it was played. The games were played in Zero Sum, a world of sixteen agents built by Ari Sklar on Softmax's Coworld platform.

The series of games. Three games to measure the slow world (section 3). One hundred games for the advisor in shapes, with three creatures on the same twenty games and twenty more games for two of them, following an extension rule written before seeing the data (section 7). Forty games for curiosity as a pull and forty for curiosity as a trip with commitment (section 8), the latter preceded by a few preliminary test games, which found four defects before spending on the series. In each series, the same seeds for all creatures: same map and same rivals in the same seats, checked game by game. The same seed always gives the same world at the start, but it does not guarantee the same game: in a short check run by Ari Sklar, five games with the same seed, each only the first twenty seconds long, one came out slightly different. It is a small check, and it should be read as such. The cause is timing, not chance in the world: every random draw in the game uses the seed, but the server advances twenty-four ticks per second of real time and applies whatever actions have arrived, so an action that arrives one tick later changes the game from that tick on (Ari Sklar, from the game's code). That is why, when we pair games by seed, we pair the same starting point, not the same game. All the games in this work were played with the same version of the game, version 0.1.18 of Zero Sum, which in its repository carries the tag zero-sum-v0.1.18 (github.com/arisklar6/battle-royal/tree/zero-sum-v0.1.18). We checked in the logs that each setting of the slow world was really applied: the duration, the twenty still seconds and the ring. We also doubled the sponsor's budget, and that changed nothing: the budget only decides whether the scheduled packages drop, each pair's packages use up eighty, and with eighty or more the same ones drop.

The house rules. Each prediction was written before playing, with the figure that had to come out, and it is reported as it came out, whether it failed or not; the ones that failed are stated in their place. Each thing is checked in the place where it really happens, after a rule of the previous work did not work without anyone seeing it (section 2) and after the report to the advisor arrived empty in the first batch of the field (section 7). The counters distinguish "does not switch on" from "switches on and changes nothing", because a zero can be a badly placed counter: it happened several times in this work and that is how it was caught. Each series is played with a single version of the program, identified by its fingerprint in the record.

What was not touched. The engine is the one from the first work, published and with the same fingerprint. The table and the body's decider are those of the previous work, without changes. Everything new, the wide door, the shapes, trust, curiosity and commitment, is placed alongside and can be switched off; with everything switched off, the body decides exactly the same as before, checked instant by instant.

The details of the door. The minimum of life that no leg can cross is 15 out of 100. In the sum along the way, what happens at fifty instants counts half as much as what happens now, and so on. A plan has to beat the body's best by a margin of 0.02 on the scale of discomfort; for the advisor, the margin grows when trust falls. Imagining a whole plan takes longer than an instant of the world, so it is done apart: with the advisor's plans, when the plan arrives and then every twenty-five instants; with curiosity trips, the first examination never before the body could have arrived; in the field, the body's decision almost never had to wait for the judgement (with the advisor's plans, in 0.2% of its instants), but some of its steps did not reach the world, as told below.

The advisor. In the field, Claude Haiku 4.5, called through the platform's intermediary; on the language bench, Haiku 4.5 and Claude Sonnet 4.5, called directly. The bench tests with a real model cost 4.44 dollars from the author's account; the games were played with the platform's credit, at no cost to the author, and their expense is given according to what the server reports. Six of two hundred complete episodes did not report their cost and have been counted as zero.

Inherited defects. Going through the diaries for the next work, we found three defects of the body that came from the previous work and that this work did not touch. The first is in the gift, the gesture of dropping a resource for the sibling to pick up: two of its rules contradict each other when the sibling is healthy, one drops the item and the other undoes the gesture, and the body drops and picks up the same thing every other instant, standing on top of it. It happened in 39 of the 379 lives of this work's series and took up 2.49% of their instants; almost all the gifts recorded in these diaries are that oscillation, not real gifts. The second: the body does not know that its backpack is full, and sometimes it keeps trying to pick up something that does not fit, because picking up and staying still tie and the tie-break rules out staying still. It happened in 182 of the 379 lives, and in the worst one, for almost half of the life. The third is a choice of the previous work whose cost we now know: for the body, a rival without a weapon is not a threat, and those

rivals also hit. They landed 705 blows and caused 30 deaths. No comparison in this work is affected, because all three were present in every arm alike, and this work does not measure the gift. All three are corrected in the next work.

Steps that did not arrive. The next work also found a defect in the machinery of this work, not in the body. In the arms with the door, while preparing the judgement that is done separately, the body copied its whole memory, and meanwhile some of its steps did not reach the world: it decided to move and stayed on the same square. With the body alone this never happened. With the advisor's plans, 6.2% of the steps towards a free square were lost; with the random shapes, 2.6%; with curiosity, 10.4%, above all while a plan was being walked. These shares are counted over whole lives; they grow towards the end of the game, which is why the next work, counting by stages, gives higher figures for the final ones. A lost step can only make an arm do worse, so the defect biases each comparison against the arm that lost more steps. Measured again only on the stretches where no step was lost, the conclusions hold, and the one about curiosity comes out stronger: the advisor's plans meet 45% of their checkpoints, against 63% for the random shapes; curiosity discovers 21% more new squares than the body alone and, in the paired comparison, sees 31% more of the world, still far from the half again the seal asked for, and while it travels it still discovers almost six times more world per instant than when it is not traveling; commitment is obeyed in 92% of the trips, although obeying meant giving the order for the step, and 90 of those orders did not reach the world. Life and placing cannot be measured this way, because a life without a single lost step tends to be a short one; about them we still say the same: no difference. Where this work said that the body never had to wait, what was measured was that no decision was skipped; whether the steps reached the world, nobody measured. In the next work the judgement runs in a separate process, and no step is lost.

What this work cannot say. It is a single world, and a private world: we removed the three types of rival that attacked most at the start of the game, and one of those we put in their place turned out to kill late (section 3). Nothing measured says how the creature would do among the hunters that attack most. The body's imagination sees the rivals as still, and the honesty rule of section 5 prevents a plan from claiming gains from them, but not the others from doing something nobody foresaw. One hundred games do not distinguish a small effect from noise in a world where chance weighs a lot; what we say does not change did not change in a way that could be seen. The bond with strangers was tested only on the bench. And the creature learns nothing from one game to another: everything it does, it does with what it already had.

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To Ari Sklar, of Softmax, who maintains Zero Sum, the world where these games were played. He answered with data every question we asked him about that world, searched the game's code for what was not written anywhere, such as the way the sponsor hands out its packages, warned us that the same seed does not guarantee an identical game, and carefully corrected what we were telling wrongly about the platform. Several sentences in this work are more accurate thanks to him.

To Softmax, for the Coworld platform, where the game and its leagues live, and for the credit with which all the games were played, which cost the author nothing.

Assistance statement

This work was done with the help of language models from Anthropic (Claude), used on three fronts: programming the agent, running and checking the experiments, and writing this text. The advisor that speaks inside the games is also one of those models, and this is stated in section 10. The same rule was applied to that collaboration as to everything else: each figure cited comes from a log and was checked against its source, and no claim rests on the model's memory. The design of G-EMV, the decisions of this work and the responsibility for everything written are the author's.

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Appendix. The numbers and where they come from

The figures in the text appear here with the exact value that was measured and the report they come from; thresholds that were fixed before measuring are marked as such. In the text the figures are rounded; here they are given in full. The reports are in the work's data folder (cantera/paper5/), each with the version of the program that produced it. All field games were played in Zero Sum 0.1.18 (tag zero-sum-v0.1.18, commit 9bd0ddf, in the repository arisklar6/battle-royal, formerly called arisklar6/zero-sum, which redirects to it; it is an annotated tag, so whoever checks it must dereference it to see that commit). The engine is the one from the first work, without changes (md5 1e511978c251130e95169ebf8443efa1).

A.1 The slow world, setting by setting

Setting sent	Value	What happened in the games	Source
max_ticks	18,240	games of 18,240 ticks (12 min 40 s), twice the default	informe_P5ARI3
freeze_ticks	480	20 s without attacks; first movement at tick 482 (moving is allowed, but stepping off the starting square kills on the spot)	informe_P5ARI3
zone.schedule	from tick 7,296, 25% slower	the ring warns at tick 7,296 (40.0%) and closes 1.25 times slower	informe_P5ARI3
sponsor.budget_per_team	600 (default 300)	no effect: each pair's scheduled packages use up 80 (20 the ration pod and 60 the medkit); any budget of 80 or more gives the same as 300; below 80 packages are lost, and with 0 none. Agents cannot ask the sponsor for anything	informe_P5ARI3; Ari Sklar's review of the code at zero-sum-v0.1.18
sponsor.live	false	read by the engine (src/zero_sum/sim.nim:212); false is the default of the competition variant, so it changed nothing. live does not turn the sponsor off: it only opens the console through which a person can give gifts live, in a local game. The "enabled" the agent sees is something else, fixed to true (obs.nim: 174)	informe_P5ARI3; Ari Sklar's review
sponsor's gifts	those of the competition variant	8 pods of two rations (announced at ticks 2,016 to 2,184, 11% of the game) and 8 medkits (4,608 to 4,776, 25%), one team every 24 ticks; they land five seconds (120 ticks) after the announcement. For the pair (team F): rations announced at tick 2,136 to seat 10, first aid at 4,728 to seat 11. These are the configuration's ticks; the diary records each event one tick later	informe_P5ARI3, informe_P5ARI4
rivals	roster_lento_v1	the three policies that attacked most at the start, replaced by zero-sum-savenger, relh-zero-sum and belobog, three times each; the pair in seats 10 and	informe_P56B, lines 51 and 60; informe_P5AP1

Setting sent	Value	What happened in the games	Source
		11. relh-zero-sum is not peaceful: in the normal game it causes 52 of 418 deaths (12.4%), all after tick 2,000	
who killed the creature in the slow world	-	389 deaths in 200 episodes (399 diaries): relh-zero-sum 156 (40.1%), the first at tick 8,191, median 8,723; also the one that landed the most hits, 1,373 (32.5%). Belobog, 0 deaths; zero-sum-scavenger, 3. The ring, 14.1%. Unattributed, 11.05%. Distribution stable across the three series	informe_P5AP2

A.2 By section

Section	In the text	Measured value	Source
1 and 2	about sixteen options when it can walk	median of 16 with the legs ready	informe_P5AP1
5	the options it already had in hand, about eighteen	median of 18 candidates in the comparison handful	informe_P53D, line 21
2	eleven instants of the legs' pause	cooldown 16 – speed = 11 ticks per square	informe_P52b; the formula, from the game's code: src/zero_sum/sim.nim:144 at the tag zero-sum-v0.1.18
2	three in four instants with the legs paused	75.69% of live ticks	informe_P52b
2	nine in ten decisions to stay put, with the legs cold	90.38% of the noop	informe_P52b
2	truly still, eight in a hundred	7.96%	informe_P52b
2	the tie rule of the previous work, four in almost 2,000	4 of 1,908 wins	informe_P56A
3	calm: from three to seven in a hundred, and eighteen	3 to 7% (normal game); 18.17% (slow world)	informe_P51_C
3	not one instant with nothing lacking, in almost 40,000	0 of 37,750 live ticks	informe_P51_C
3	hands well supplied, a little more than one in a hundred	1.24% of ticks (7,960), in 10 of 260 diaries	informe_P51_E, line 81
3	four new squares in calm, fourteen outside it	4.13 and 14.10 per hundred ticks	informe_P51_C

Section	In the text	Measured value	Source
3	in the first 2,000 instants, three of fourteen types of rival: nine in ten hits and eight in ten deaths	3,612 of 3,928 hits (91.96%) and 174 of 211 deaths (82.46%) before tick 2,000; over all deaths in the 434 diaries, 190 of 418 (45.45%)	informe_P51_E, lines 154 to 186 (the 92% of deaths on line 188 is an error in the report); informe_P5AP1
4	more than 2,200 moments	2,230 scenes from 40 diaries	informe_P52c
4	good 8.5 in a hundred, silly seven	8.52% and 7.22% (n = 845 of the 908 moments with a better future: in the other 63 the good advice was not to move, and the usual door cannot judge a destination that is the body's own square; the random advice is compared on the same 845 scenes)	informe_P52c
4	exposure signs four in ten rejections	S-8-EXPOSICION, 40.51%	informe_P52c
4	one step away, two in a hundred; three or more away, twenty-three	2.35% and 22.96%	informe_P52c
4	in fourteen in a hundred, the best was not to move	125 of 908 scenes (13.77%)	informe_P52c
5	the good one seventy-one in a hundred, the silly one twenty-four points less	70.88% and 46.79%	informe_P53B
5	the imagination gets it right in nine in ten of 180 trips	zero error at the median and at the 90th percentile, at 100 ticks, in 180 windows	informe_P53A
5	with the honesty rule, one in twenty gets to be judged	4.52%	informe_P53H
5	of those judged, eighty-five good and forty-three at random	85.37% and 42.86%	informe_P53H
5	nine in ten parts go through the others	87.4% (normal game); 90.1% (slow world)	informe_P53F
1 and 5	almost half of that is the sibling; half of everything, the strangers	of a total improvement of 212.3: sibling 85.3 (40%), rivals 104.6 (49%)	informe_P53F
5		51.2% of its contribution	informe_P53H

Section	In the text	Measured value	Source
	the sibling tells where it will be: the door recovers half		
6	the advisor that speaks at random loses trust	34 of 40 lives below 0.2 (P5-4A); 39 of 40 (P5-4B)	informe_P54A, informe_P54B
6	half trust, the worst	with C = 0.5 the difference good against random is +0.02	informe_P54A
6	more than 3,000 answers; it talks about the others eight in ten	3,284 answers; 80.9%	informe_P55A
6	half the time against almost nine in ten	arrivals 49.0% against 87.5% for "nothing changes"	informe_P55A
6	loose steps: none of 367; shapes: one in ten; large model, almost two	0 of 367; 10.48%; 18.13%	informe_P55B
6	on the hands it gets two in ten right; on the bond, six	20.08%; 60.23%	informe_P55B
6	names the rivals one in seven (before, seven in ten)	14.5% and 69.5%	informe_P55B
6	stays silent one or two in a hundred	1 to 2%	informe_P55B
7	one hundred games, two hundred lives, 1.5 million instants	200 diaries; 1,501,210 live ticks; 0 decisions skipped (does not measure whether steps reached the world: row on steps that did not arrive, section 10)	informe_P56C
7	more than 2,000 consultations, all with their report	2,088 of 2,088	informe_P56C
7	two or three plans accepted in a hundred	2.4% with the advisor (arm F); 1.7% with random shapes (arm T)	informe_P56C
7	more than 3,000 plans rejected; the body alone was not better off	3,568 shapes rejected by area	informe_P56C
7	three in four steps, the plan's	76.8% with the advisor, with the legs ready	informe_P58O
7	the advisor's plans met forty-two in a hundred stops	42.3% per checkpoint with the advisor, against 61.1% with random shapes	informe_P56C, informe_P58O
7		40.8% of the calls	informe_P56C

Section	In the text	Measured value	Source
	it stayed silent four in ten		
8	more than 6,000 instants of calm in a row without looking	ticks 1,977 to 8,027 (6,050), map left to see fixed at 0.7969; life P57C_t2_A_20994019, seat 10 (body alone, final slow world)	figuras_paper5/fig3_meseta_de_calma.txt
8	without the balance, ten times more changes in unsafe	from 0.48% to 5.39%	informe_P57A
8	the unseen, six squares away	median of 6 squares (never more than 11)	informe_P58A, line 91
8	the pull, in safe surroundings, changes one decision in eighteen	5.62%	informe_P57C, line 187
8	the pull in the field: 9% more map	1.086 times; 14.5 new squares per hundred ticks in both arms	informe_P57C
8	without the relief, three in a hundred trips; with it, one in three	2.68% and 37.25%	informe_P58A
8	obeys nine in ten strides	92.5%	informe_P58M
8	seven in ten trips are completed; none steps on its destination	68.9% (144 of 209); completed = sees the promised drop in map or the destination square; destination stepped on, 0 of 209	informe_P58M; informe_P5FIG2
8	the body alone goes back and forth between two squares during the calm	squares (19,13) and (18,12), 527 moves, one every 11 ticks, from tick 2,033 to 8,027; two options take turns: from (19,13) going for some arrows at (18,15) wins, and its first step is (18,12); from (18,12) that option is not generated and going for loot wins, which returns to (19,13); the arrows are never picked up. From (19,13) it sees a rival standing still and from (18,12) it does not: discomfort 3.6767 and 3.26065	informe_P5FIG2
8	almost six times more world inside a trip	5.81 times, in the 20 seeds; 5.58 on the stretches with no lost step (95% CI by lives 4.2 to 7.2)	informe_P58M, informe_P5ARI4
8	five causes of breaking	5 of the 7 foreseen breaks fired	informe_P58M
8	the seal asked for half again as much map	threshold fixed before measuring: 1.5 times	informe_P58M
8	sees 16% more	1.16 paired (10, 15 and 20 seeds)	informe_P58M
8		2.2%	informe_P58M

Section	In the text	Measured value	Source
	the shape lives 2% of the life		
8	one shape every forty-three safe instants	0.0205 to 0.0313 shapes per safe tick; from 10 to 383 shapes per game	informe_P58M
8	costs neither life nor place	life p = 0.50; place p = 1.00	informe_P58M
8	four hundred thirty-four lives; no death at the hands of an unknown one (never seen)	0 of 418 deaths at the hands of a seat with zero knowledge	informe_P57B
8	knows more than nine tenths about its killer	knowledge of the killer, median 0.94	informe_P57B
8	the bond changed some decision with an armed rival within range, one or two	1 decision with k = 0.1 and 2 with k = 0.2, in two configurations that never ran together	informe_P57B
8	with curiosity as a pull, more than 27,000 decisions with an armed rival within range, none of curiosity	0 of 27,519 (series of the pull)	informe_P57C
8	two predictions failed because of how they were measured	armed rival within range with an active trip: 52 ticks, 5 with legs ready (breaks by veto); cost per tick: one seat of 39 at 5.51 ms in all its ticks, zero decisions skipped	informe_P58M
10	inherited defect 1, the gift: drops and picks up the same thing	74,545 ticks in 379 lives (2.49%); 39 lives affected; 95.1% of the 4,576 recorded gifts were the oscillation	informe_P6_6 (cantera/paper6/)
10	inherited defect 2: tries to pick up with a full backpack	47,244 ticks in 182 of 379 lives; worst streak 5,593 ticks (44% of that life); picking up ties with staying still and the tie-break rules out staying still	informe_P6_8 (cantera/paper6/)
10	inherited defect 3: an unarmed rival is not counted as a threat	705 blows and 30 deaths from visible rivals with an empty hand	informe_P6_8 (cantera/paper6/)
10	steps that did not reach the world, in the arms with the door	ready steps towards a free square not executed: A 0%, F 6.2%, T 2.6%, K 10.4%; on clean stretches: checkpoints F 45.1% (n 51) against T 63.0% (n 27), difference -17.9 [-38.0, +5.2]; paired 48.5% against 60.0%; new squares K/A 1.213; paired world 1.308; commitment obeyed 91.9% in	informe_P6_22 (cantera/paper6/), informe_P5_LAG (cantera/paper5/)

Section	In the text	Measured value	Source
		168 clean shapes of 209, 90 issued steps not executed; life and placing cannot be restricted (lives without a lost step are short); decisions that waited for the judgement, F 0.215%	
10	bench tests with a real model	603 calls, 4.44 dollars	informe_P55B
10	six of two hundred episodes without reported cost	6 episodes, counted as zero	informe_P5ARI_cierre

A.3 What it cost

The recorded cost of paper five was 31.33 dollars; six episodes did not report their cost and were counted as zero, so the real expense may be somewhat higher. Of that, 16.08 went to games on the platform, with its credit, and 10.81 to the advisor called through the platform, also with its credit: neither of the two cost the author anything. The remaining 4.44 went to the bench tests with a real model, paid for by the author. By series: the advisor in shapes, 7.87 of platform and 10.48 of model; curiosity as a pull, 2.67; curiosity as a trip, 3.44; and ten other series and loose tests, 2.43 among them all. The full breakdown is in informe_P5AP1.

A.4 The figures

Each figure was drawn with cantera/paper5/fig_paper5.py from the diary of one or several lives. Its record, with the exact data and the reason that life was chosen, is in the folder figuras_paper5/.

Figure	Which life	Measured values	Record
1	P5-7C, batch 2, arm A (body alone), seed 20994019, seat 10	10,752 ticks lived; first blow at tick 10,273 (7.1 min); 24 blows in the whole life; what it lacks never drops below 0.6667; place 13; the solid band in the bottom panel is the back and forth between two squares (ticks 2,033 to 8,027): the rows F-4-ALCANCE and S-8-EXPOSICION switch on and off every 22 ticks. Chosen among the 93 arm A lives with more than 6,000 ticks	fig1_una_vida.txt
2	P5-6C, batch 5, arm F (body with advisor), seed 22250767, seat 10; shape 7, born at tick 3,999 and accepted at 4,001	checkpoints at ticks 4,004, 4,037, 4,092 and 4,092; dropped at tick 4,062 because it stops winning. The imagined curves are not in the diary: they are recomputed with the same door code and reproduce the recorded advantage (0.28435) and the checkpoints. Best case: 10 of the 45 arm F shapes step on the destination of their first leg, and this is one of them; the destination of leg 2, (22,22), was not stepped on within the plan. At the first checkpoint, reality comes out 0.519 better than imagined, and 0.500 of that is the rival's floors imposed by the honesty rule (informe_P5FIG2)	fig2_forma_de_un_plan.txt
3	the same life as figure 1	ticks 1,977 to 8,027 (6,050) with zero threat and map left to see fixed at 0.7969	fig3_meseta_de_calma.txt

Figure	Which life	Measured values	Record
4	P5-8M, batch 0, arm K (curious body), seed 20260916, seat 11: the seat with the most accepted trips in the forty games	18 trips: 5 completed and 13 stuck (three steps without getting closer to the destination); the 13 in a row go to the same destination, (21,29); none steps on its destination. No trips from tick 4,907 to 8,337 (threat below 0.2 in only 9 of 3,431 ticks) nor after (17 proposed, all 17 turned down by the door; top advantage 0.00455 against a margin of 0.02); informe_P5FIG2	fig4_una_vida_curiosa.txt
5	P5-8M, the forty games, seats 10 and 11 of both arms	40 lives of the body alone and 39 of the curious body (the diary of seat 10 in game 20784561 is missing, informe_P5ARI_cierre); map left to see rebuilt every 200 ticks with the same piece the body uses; median drawn only with 5 lives or more	fig5_cuanto_mundo_conoce.txt